Alg. number theory
Serie 4
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Points 3.75/18 (20.83%)

Aufg 1.)
Let $F = \mathbb{F}_2(T)$, $K = F(\sqrt{T})$

i.) t.s. K/F is a field extension with $[K:F]=2$.

proof By construction of K, the first statement follows. ok
Then $\min_{K/F}(\sqrt{T}) = X^2 - T \in \mathbb{F}_2(T)[X]$
(since X^2-T is irreducible by eisenstein crit with $p = T$, ??
and is monic and $\sqrt{T}$ a root of it.
Thus $[K:F] = \deg(\min_{K/F}(\sqrt{T})) = 2$. ✓ 0.5T

ii)
t.s. $\mathrm{Tr}_{K/F}(x) = 0 \ \forall x \in K$

proof Let $x \in K = F(\sqrt{T})$.
By i) it holds that $2 = [K:F] = [K:F(x)] \cdot [F(x):F]$.
Thus we have two cases.
- If $[K:F(x)]=2$, then by Lemma 1.8 we have
$$\mathrm{Tr}_{K/F}(x) = -a_1 \underbrace{[K:F(x)]}_{=2} = 0 \text{ in } \mathbb{F}_2(T) \text{ where } a_1 \in \mathbb{F}_2(T).$$ ok
- If $[K:F(x)]=1$, then $[F(x):F]=2$
Since $[K:F]=2$, we have $\min_{K/F}(x) = X^2 + a_1 X + a_0 \in \mathbb{F}_2(T)$.
But then $\dim_F(K)=2$ thus $(1,\sqrt{T})$ is a basis of K hence
there exist $b_1, b_2 \in \mathbb{F}_2(T)$ s.t.
$$x = b_1 + b_2\sqrt{T}$$
Then $x^2 = b_2 T + \underbrace{2 b_1 b_2 \sqrt{T}}_{=0 \text{ in } \mathbb{F}_2(T)} + b_1^2$
Thus $a_1 = 0$ and $a_0 = -(b_2 T + b_1^2)$ and by Lemma 1.8.
$\mathrm{Tr}_{K/F}(x) = 0$. ✓ 2

iii) t.s. $\forall x_1, x_2 \in K : \Delta_{K/F}(x_1, x_2) = 0$

proof Let $x_1, x_2 \in K$.
By def $\Delta_{K/F}(x_1,x_2) = \det\left((\mathrm{Tr}_{K/F}(x_i x_j))_{i,j \in \{1,2\}}\right)$
But with ii) every entry 0 thus
$\Delta_{K/F}(x_1,x_2) = 0$ ✓ 1 ✓